Luké de Beer

Software Developer · Final-year Computer Science Student

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SUMMARY

Final-year Computer Science student at Stellenbosch University, graduating in December 2026, and technical co-founder of Responsagility, a behavioural-acuity platform currently being used in production. I built the platform from the ground up, covering everything from the database and backend to authentication, testing, deployment, and reporting.

I enjoy working across the full stack and solving problems that require both careful engineering and practical thinking. I also make extensive use of AI in my development process, but focus heavily on testing and review to make sure the resulting code is reliable and maintainable. I work well independently, communicate clearly, and am comfortable taking ownership when a project needs someone to drive it forward.

EXPERIENCE

Technical Co-founder & Developer
Responsagility · responsagility.me **January 2026 – Present · Somerset West**

Web platform (live in production)

- Designed and built the web platform from the ground up, including the frontend, backend, database, authentication, testing, deployment, and reporting.
- Built a multi-tenant system that allows multiple organisations to use the platform while keeping their data strictly separated.
- Added multiple layers of data protection, including PostgreSQL row-level security, organisation-level access checks, and validation at API boundaries.
- Built an AI-assisted content pipeline for creating and reviewing behavioural simulations, while keeping the scoring system deterministic and independent from the AI.
- Added tools for personal-data deletion, branded PDF reports, and Excel exports to support real-world client requirements.
- Built and maintain a 1,500+ case automated test suite covering both application logic and PostgreSQL integration.
- Manage releases through a migration-first workflow to reduce the risk of database and deployment issues.

Mobile app (iOS / Android)

- Built a cross-platform daily-practice app from the ground up, including authentication, account management, backend services, and AI-generated reflections.
- Implemented registration, email verification, password recovery through deep links, and account deletion in line with App Store requirements.
- Designed the submission system to safely handle retries, offline states, and fresh installs without creating duplicate entries.
- Integrated AI-generated daily and weekly reflections with scheduled processing and a shared persona system to keep the experience consistent.
- Secured the backend using JWT verification and user-level access controls.

Freelance AI Evaluator
Outlier.AI **October 2024 – February 2025 · Remote**

- Reviewed and tested AI-generated answers and code to identify errors, edge cases, weak reasoning, and other problems.
- Provided detailed feedback used to improve the quality and reliability of AI models.

Job Shadowing
Stone Three & Mediclinic **2019 · Somerset West**

- Gained early exposure to professional software development through Stone Three and to healthcare operations through Mediclinic.

UNIVERSITY PROJECTS

OAuth 2.0 / OpenID Connect Server · Spatialedge
Software Engineering (CS344) · team of 6 **July – October 2026**

Built the core of an OAuth 2.0 / OpenID Connect sign-in server for Spatialedge without using an OAuth library, along with a notes app that signs in through it. My work covered the token, consent and signing-key endpoints, two-factor login and the production deployment, and I wrote the largest share of the team's code and tests. TypeScript, React, Node.js, SQLite, Docker.

Computer Networks Projects

Computer Networks (CS313) · team of 4 **February – May 2026**

Four networked applications in Java: a multi-client chat app, a reliable UDP file-transfer protocol, a VoIP app with group calls, and an anonymous peer-to-peer file-sharing system. I wrote the chat server, both ends of the file-transfer protocol and the VoIP client, and added RSA and AES-GCM encryption to the file-sharing system.

Parallel Othello AI & Process Scheduler

Concurrency (CS314) · individual **February – May 2025**

An Othello-playing agent in C that uses MPI to search moves in parallel with minimax and alpha-beta pruning, guided by an evaluation function that adapts as the game progresses. Also built a process scheduler simulator with first-come-first-served, round-robin and priority scheduling.

EDUCATION

BSc Computer Science · Stellenbosch University **2023 – 2026 · Graduating December 2026**

Marketing & Management Sciences · IMM Graduate School · First year completed **2022**

Mechatronic Engineering · Stellenbosch University · Transferred to Computer Science **2021**

Matric · Paul Roos Gymnasium **2020**

INVOLVEMENT

Volunteer · Aitsa! After Care Centre · Assisted with activities and supported daily operations **2018**

SKILLS

Languages: TypeScript, JavaScript, Python, C, Java, SQL

Frameworks & Development: React, Next.js, React Native, Expo, Node.js, Express, Fastify, JavaFX, MPI

Databases & Infrastructure: PostgreSQL, Supabase, SQLite, Drizzle ORM, Docker, Caddy, Railway, GitLab CI, Git

Security & Networking: OAuth 2.0 / OpenID Connect, JWT, TOTP two-factor authentication, RSA and AES-GCM encryption, TCP/UDP socket programming

Testing: Vitest, React Testing Library, JUnit, Unit testing, Integration testing

AI-Assisted Development: AI-integrated development workflows, Anthropic and OpenAI APIs, custom Claude Code workflows